ADVANCED INTELLIGENT DISCOVERY
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CALL FOR PAPERS / SPECIAL ISSUE

Self-Driving Labs: Autonomous Platforms for Scientific Discovery

Advanced Intelligent Discovery

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Overview

Self-driving laboratories (SDLs) are redefining the practice of scientific research by integrating robotics, automation, artificial intelligence, machine learning, and advanced data infrastructure into closed-loop discovery platforms capable of designing, executing, and interpreting experiments with minimal human intervention.

Over the past decade, SDLs have evolved from proof-of-concept systems into increasingly sophisticated platforms operating across chemistry, materials science, biotechnology, and pharmaceutical discovery. These systems promise not only to accelerate discovery but also to improve reproducibility, accessibility, and efficiency while enabling new modes of scientific exploration.

This Special Issue of Advanced Intelligent Discovery seeks to bring together researchers developing and deploying autonomous experimental platforms as well as the enabling algorithms, software, infrastructure, and community frameworks that support them.

Topics of Interest

Submissions may address, but are not limited to:

Autonomous synthesis and robotic platforms

- Materials discovery platforms
- Molecular synthesis systems
- Flow and continuous chemistry
- Autonomous characterization and testing
- Mobile laboratory robotics

AI-driven experiment design

- Bayesian optimization
- Active learning
- Reinforcement learning
- Multi-objective optimization
- Autonomous decision-making systems

Data management and infrastructure

- Workflow orchestration
- Laboratory operating systems
- Scientific workflow management
- Data standards and interoperability
- Knowledge graphs and scientific databases

Biological and pharmaceutical SDLs

- Drug discovery platforms
- Synthetic biology workflows
- Protein engineering
- Bioprocess optimization
- Automated biological experimentation

Hardware and laboratory automation

- Modular robotics
- Laboratory instrumentation
- Sensor integration
- Experimental automation architectures
- Human-robot collaboration

Benchmarking and community infrastructure

- Reproducibility and validation
- Cross-laboratory benchmarking
- Shared datasets and open platforms
- Standards development
- Accessibility and scalability

Article Types

The Special Issue welcomes:

- Original Research Articles
- Review Articles
- Perspectives
- Outlooks and Community Papers

Important Dates

MILESTONE	DATE
Call announced	**November 2026**
Submission deadline	**1 July 2027**
Publication	**Online upon acceptance**

Why Contribute?

This Special Issue aims to provide a comprehensive snapshot of the rapidly evolving SDL landscape and to capture the scientific, technological, and organizational advances that will shape the next generation of autonomous scientific discovery.

Particular emphasis will be placed on contributions that address:

- Reproducibility and robustness
- Accessibility and democratization of SDLs
- Integration of AI and laboratory automation
- Scalable and deployable autonomous systems
- Community frameworks, standards, and shared infrastructure

Researchers from academia, national laboratories, research institutes, and industry are all encouraged to submit.

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How to Submit

Manuscripts should be submitted through **https://authors.wiley.com/journal/AIDI**, selecting this Special Issue. Prospective authors are welcome to send a short title and abstract to the Guest Editors beforehand to check fit.

Advanced Intelligent Discovery

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